OCR Further Mathematics
Statistics - Continuous Random Variables
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245

The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a		f(x) = ½ $\int_0^2 \frac{1}{2} a\cos(ax)\mathrm{d}x = 0.3$ $\left[\frac{1}{2}\sin(ax)\right]_0^2$ $\frac{1}{2}\sin(2a) = 0.3$ a = 0.32175…	B1 (AO3.3) M1 (AO3.1a) B1 (AO1.1) M1 (AO2.1) A1 (AO1.1) [5]	Stated or implied, e.g. on diagram $\int \mathrm{f}(x)\, a \cos ax\, \mathrm{d}x$ & equated to 0.3 Correct indefinite integral Correct limits, solve Answer, a.r.t. 0.322 (ignore other answers)	
	b		$\mathrm{F}(y) = \frac{1}{2}y \quad [0 \le y \le 2]$ $\mathrm{P}(Y^2 \le m) = \mathrm{P}(0 < Y \le \sqrt{m})$ $= \mathrm{F}(\sqrt{m}) \quad [= \frac{1}{2}\sqrt{m}]$ $\frac{1}{2}\sqrt{P_{60}} = 0.6$ $P60 = 1.44$	M1 (AO3.1a) A1 (AO1.1) M1 (AO2.1) A1 (AO1.1) M1 (AO1.1) A1 (AO2.2a) [6]	Use their f(y) to obtain CDF Correct F(y) (range need not be stated explicitly) Find CDF of Y^2, allow m^2 instead of $\sqrt{m}$, or $\pm\sqrt{m}$, here Use F(y) correctly Equate to 0.6 and solve, need $\sqrt{m}$ here 1.44 or exact equivalent	
			Total	11		
2	a		$\int_1^\infty kx^{-n}\mathrm{d}x = \left[\frac{k}{(1-n)x^{n-1}}\right]_1^\infty$ $= \frac{k}{n-1} = 1$ so $k = n - 1$	M1 (AO1.1) B1 (AO1.1) A1 (AO1.1) [3]	Integral attempted, correct limits Correct indefinite integral Correctly obtain $k = n - 1$, www	Don't need full details of lim($a \to \infty$)

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	b	i	$\int 3x^{-4}\mathrm{d}x = -\frac{1}{x^3} + c$	M1 (AO1.1)	Needs + c or definite integral between 1 and x, oe	Wrong k: can get M1A0B1
			$x = 1$, F(x) = 0 so $c = 1$. Hence $1 - x^{-3}$.	A1 (AO1.1)	Fully correct active part of CDF	Ignore ranges here
			$\mathrm{F}(x) = \begin{cases} 0 & x < 1, \\ 1 - \frac{1}{x^3} & x \geq 1 \end{cases}$	B1 (AO1.1) [3]	"0 for x < 1" stated and no wrong ranges (doesn't need M1 or A1) Allow ≤ for <, and / or > for ≥	Or "0 otherwise" if "x ≥ 1" stated in active part
		ii	$\frac{\mathrm{P}[(X>7)\cap(X>5)]}{\mathrm{P}(X>5)} = \frac{\mathrm{P}(X>7)}{\mathrm{P}(X>5)}$	M1* (AO3.1a) A1 (AO3.1a)	Use conditional probability method P[(X > 7) ∩ (X > 5)] = P(X > 7)	$\frac{[1-\mathrm{F}(7)][1-\mathrm{F}(5)]}{1-\mathrm{F}(5)}$:
			$= \frac{1-\mathrm{F}(7)}{1-\mathrm{F}(5)}$	*depM1 (AO3.3)	Convert probabilities into F(X), *not* using P(X > 7) × P(X > 5)	M1A0M0A0
			$= \frac{125}{343}$ or 0.364(431...)	A1ft (AO1.1) [4]	Any exact fraction or awrt 0.364, ft on $1 - a/x^3$, $a \neq 0, 1$	Allow from F(x) = $1 - a/x^3$, otherwise www

Question			Answer/Indicative content	Marks	Part marks and guidance		
	c		$E(X^2)=\int_1^{\infty} kx^{2-n}\,dx=\left[\frac{kx^{3-n}}{(3-n)}\right]_1^{\infty}$ $(n\neq 3)$	M1* (AO2.1) B1 (AO1.1)	Correct limits needed somewhere Correct indefinite integral or $\frac{n-1}{n-3}$	S $E(X^2)=\frac{n-1}{n-3}$, C: M1B1 $E(X)=\frac{n-1}{n-2}\Rightarrow n\neq 2$ or 3:	
			If n = 3, $E(X^2)=\lim_{x\to\infty}[2\ln(x)]$, not defined				
			Infinite integral does not converge if 3 – n ≥ 0	*depM1 (AO2.2a)	No marks just for this unless last 3 marks all zero, then if this (or for n = 2) is shown, award SC B1	(not valid, must consider ln if n = 2 or 3): B0	
			If n ≥ 4 $E(X)=\left[\frac{kx^{2-n}}{(2-n)}\right]_1^{\infty}$ converges then	B1 (AO2.3)	Make deduction based on convergence, ft	No limits used: M0B1M0B0	
			Therefore Var(X) is not defined if and only if n = 2 or 3.	A1 (AO2.2a) [5]	Consider convergence of E(X) Shown not defined for n = 2 or 3 *and only* for those	SC: Var(X) < 0 when n < 3: M1B1M1 (B0) A0 But no need to state "if and only if"	
			Total	15			

Question			Answer/Indicative content	Marks	Part marks and guidance	
3	a		Any reason for independence (or not) … and for constant average rate (or not), in each case without misunderstanding of what they mean	B1 (AO3.5b) B1 (AO3.5b) [2]	"Events occur independently and at constant average rate": B0 SC: Mere assertion of both, properly contextualised: B1 SC: Variance = 4.67 which is closer to 5: B1 SC: Considers only the assumptions given in the B0	
	b	i	0.146(223) BC	M1 (AO3.4) A1 (AO1.1) [2]	Correct method stated or implied Correct answer only, awrt 0.146	
		ii	0.133(372) BC	M1 (AO1.1) A1 (AO1.1) [2]	0.068: M1A0 (treat 0.1337 as a slip, i.e. give A1 BOD)	
	c		Po(12.2) P(≤ 15) – P(≤ 9) [= 0.8296 – 0.2253] = 0.604(224) BC	M1 (AO3.3) M1 (AO1.1) A1 (AO3.4) [3]	Stated or implied Allow P(≤ 16) or P(≤ 10), e.g. 0.503 or 0.662 (M1M1A0) Allow this M1 also from λ = 7.2 (0.187, 0.110, 0.189) Correct answer only, awrt 0.604	
	d		Sales of CD players and integrated systems need to be independent	B1 (AO1.1) [1]	Need "independent" or "not related" clearly referred to the two types of machine. *Not just* "purchases independent" or "distributions independent"	

Question			Answer/Indicative content	Marks	Part marks and guidance	
	e		If a customer buys a CD player they probably won't (or will) buy an integrated system as well α: May buy both so not independent: B0 β: Often bought together: B1 γ: Misunderstanding of context, e.g. CDs / CD players, or assuming that integrated systems don't include CD players: can get B1	B1 (AO3.5b) [1]	Any reason for non-independence of sales of CD players and integrated sound systems Can get B0B1 provided they are focussing on independence	
			Total	11		
4			$53.1 \pm 1.96\sqrt{\frac{30}{8}}$ (49.30, 56.90)	M1 (AO3.3) A1 (AO1.1) A1 (AO1.1) A1 (AO3.4) [4]	Correct structure with 8 Square root correct Awrt 1.96 used, can be implied Both, only these numbers (4 sf needed at least once)	Allow e.g. (49.30, 56.9)
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance		
5	a		Let $H(x)$ be the CDF of $2T$. Then $\mathrm{H}(x) = \mathrm{P}(X \le x) = \mathrm{P}(2T \le x)$ $= \mathrm{P}(T \le ½x) = \mathrm{F}(½x)$ $= 1 - e^{-0.125x}$ [for $x \ge 0$, and 0 for $x < 0$]	M1(AO3.1a) M1(AO1.1a) A1(AO1.1) [3]	Convert to $\mathrm{P}(2T \le x)$ Rearrange to get $\mathrm{P}(T \le \mathrm{f}(x))$ Any letter. Correct answer only, ignore other ranges Examiner's Comments Although the calculation of CDFs of related random variables is a standard syllabus item that has appeared in the specimen and practice papers (and on the old S3), many candidates did not choose the right method for this question (perhaps because it appears to be very simple). Many replaced x by $2T$, or doubled or halved the given formula. The correct method is to follow the standard procedure of saying that $\mathrm{P}(X < x) = \mathrm{P}(2T < x) = \mathrm{P}(T < ½x) = \mathrm{F}(½x)$. Here the method of using F of the inverse function also works, but that method is not recommended in general as it needs awkward adaptation for decreasing functions.	Alternatively: $\mathrm{g}(T) = 2T$, $\mathrm{F}(\mathrm{g}^{-1}(x))$: M2 Needn't be simplified	
	b		Due to the error on the paper, all candidates get 7 marks for Q9(b). Annotate each answer with SEEN and enter 7 marks in RM. The only instance where full marks would not be awarded is where a candidate has not attempted any question.	M1(AO2.1 M1(AO1.	Stated or implied		

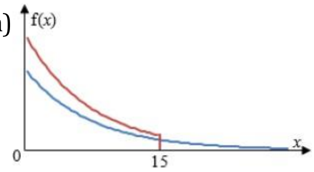
Question			Answer/Indicative content	Marks	Part marks and guidance	
			This would then need to be a 0. PDF is $f(x) = 0.25e^{-0.25t}$ $E(e^{kt}) = \int_0^\infty 0.25e^{kt}e^{-0.25t}dt$ $\int_0^N 0.25e^{-(0.25-k)t}dt = \left[-\frac{0.25e^{-(0.25-k)t}}{0.25-k}\right]_0^N$ $= -\frac{0.25e^{-(0.25-k)N}}{0.25-k} + \frac{0.25}{0.25-k}$ The first term will only converge for $k < 0.25$ The $\lim_{N\to\infty} \frac{0.25e^{-(0.25-k)N}}{0.25-k} = 0$ n $\int_0^\infty 0.25e^{-(0.25-k)t}dt$ $= \lim_{N\to\infty}\left\{-\frac{0.25e^{-(0.25-k)N}}{0.25-k} + \frac{0.25}{0.25-k}\right\}$ $= \frac{0.25}{0.25-k}$ or $\frac{1}{1-4k}$ AG	1a) M1(AO2.1) A1(AO1.1) B1(AO2.4) B1(AO2.1) A1(AO2.2a) [7]	Attempt $\int e^{kt}f(t)dt$, correct limits Method for integration Correct indefinite integral with finite upper limit Consider range of k for which the result is valid Correctly obtain given answer <u>Examiner's Comments</u> We are sorry for a misprint on this question (t should have been T) which made the question unanswerable. After analysing candidate performance during marking, we decided the fairest approach was to award all candidates full marks for Question 9b.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
	c		P(no event between 0 and θ) = P($T > \theta$) $= e^{-0.25\theta}$ P(0) from Po(λ) = $e^{-\lambda}$ Hence same expression, with $\lambda = 0.25\theta$.	M1(AO2.1) A1(AO1.1) B1(AO1.1) A1(AO2.2a) [4]	Correct method for probability Correct formula Simplified, any λ Correctly justify required result, with $\lambda = 0.25\theta$ oe stated explicitly <u>Examiner's Comments</u> This was a challenging last question, but one that should not have been entirely unfamiliar; the Specification in Section 5.03a includes 'understanding informally the link between the exponential and Poisson distributions'. Few correct answers were seen. Unexpectedly, the main reason for candidates' difficulty was that they found the probability that T was between 0 and θ; this is the probability that <u>at least one</u> event occurs between times 0 and ϑ. What was needed for the probability that <u>no</u> event occurs in this interval, which is simply $e^{-0.25\vartheta}$. Some were unable to write down a simplified expression for P(0) from Po(λ), but those who did both correctly had no difficulty in completing the question.	$1 - e^{-0.25\theta}$: M1A0 i.e. neither 0! nor e^0 left in Need to say "same" oe
			Total	14		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6		a	$\int_0^a x\frac{2x}{a^2}dx = 4$ $\left[\frac{2x^3}{3a^2}\right] = 4$ $\frac{2}{3}a = 4 \Rightarrow a = 6$	M1 (AO 3.1a) B1 (AO 1.1) A1 (AO 2.2a) [3]		
		b	$F(x) = \frac{x^2}{36}$ Let the CDF of M be H(m). Then H(m) = P(all observations less than m) = $[P(X \le m)]^5$ $= \left[\frac{m^2}{36}\right]^5$ $H(m) = \begin{cases} 0 & m < 0, \\ \frac{m^{10}}{60466176} & 0 \le m \le 6, \\ 1 & m > 6. \end{cases}$	M1 (AO 1.1) A1ft (AO 1.1) M1 (AO 2.1) M1 (AO 3.1a) A1 (AO 2.2a) A1 (AO 2.1) A1 (AO 2.1) A1 (AO 1.2) [8]	Fin $F(x);=\frac{x^2}{a^2}$ d Correct basis for CDF of m Correct function, any letter Range $0 \le m \le 6$ Letter not x, and 0, 1 present	ft on their a Allow </≤ throughout
			Total	11		

Exam Papers Practice

Question			Answer/Indicative content	Marks	Part marks and guidance	
7		a	DR $\int_0^{\infty} x\lambda e^{-\lambda x} dx$ $= \left[-xe^{-\lambda x}\right]_0^{\infty} - \int_0^{\infty} -e^{-\lambda x} dx$ $= \lim_{k\to\infty}\left(\left[-xe^{-\lambda x} - \frac{1}{\lambda}e^{-\lambda x}\right]_0^k\right)$ $= \frac{1}{\lambda}$ so $\lambda = \frac{1}{5.0} = 0.2$	M1 (AO 1.1a) M1 (AO 1.1) E1 (AO 2.4) A1 (AO 1.1) [4]	$\int_{20}^{\infty} 0.2e^{-0.2x} dx$ Use Parts used explicitly, $e^{-\lambda}$ integrated or equivalent consideration of limits Correctly obtain $\frac{1}{\lambda}$ and λ = 0.2, www	accept $= 0 - \left[\frac{1}{0.2}e^{-0.2x}\right]_0^{\infty}$
		b	$\int_{20}^{\infty} 0.2e^{-0.2x} dx$ = 0.0183156…	M1 (AO 3.1b) A1 (AO 3.4) [2]	Or $1 - \int_0^{20} 0.2e^{-0.2x} dx$ BC Awrt 0.0183	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	(a)  (b)	B1 (AO 1.1) [1] M1 (AO 3.5c) A1 (AO 3.3) [2]	Correct shape for $y = \lambda e^{-\lambda x}$, asymptotic to x-axis only Any curve truncated at 15 More than half above their curve from part (i)	Don't need vertical line i.e., awareness that total area is the same	
			Total	9			

Question			Answer/Indicative content	Marks	Part marks and guidance	
8		a	$\int_0^2 \frac{1}{4}x(4-x^2)\mathrm{d}x = 1$ $f(x) \geq 0$ for $0 \leq x \leq 2$	M1 (AO 3.1a) A1 (AO 2.2a) B1 (AO 2.4) [3]	Attempt to show this Algebra needed for "show that" Explicitly stated – not just diagram drawn	"By calculator": M1A0
		b	$\int_0^a \frac{1}{4}x(4-x^2)\mathrm{d}x = \frac{15}{16}$ $\left[\frac{x^2}{2} - \frac{x^4}{16}\right]_0^a = \frac{15}{16}$ $a^4 - 8a^2 + 15 = 0$ $a^2 = 3$ or 5 $a \neq -\sqrt{3}$ or $\pm\sqrt{5}$ (outside range) $a = \sqrt{3}$ and no other solutions	M1 (AO 3.1b) A1 (AO 1.1) M1 (AO 2.1) M1 (AO 3.1a) A1 (AO 2.3) A1 (AO 2.2a) [6]	Integral, correct limits somewhere, and equate to $\frac{15}{16}$ Correct indefinite integral Reduce to quadratic Solve for a^2 Explicitly rejected Needn't obtain previous A1	BC No reason needed
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9		a	(i) 20	B1 (AO 1.2) [1]		
			(ii) 16	B1 (AO 1.2) [1]		
		b	N(20, 16) and P($X < x$) = 0.98 [= 28.7] n = 29	M1 (AO 3.3) A1 (AO 1.1) A1 (AO 3.2a) [3]	Correct calc^{n} seen or implied 28.7 seen or implied Allow $n \geq 29$	
		c	B(100, 0.2): P(≥ 29) = 0.2002, P(≥ 30) = 0.01125 So correct answer is n = 30	M1 (AO 1.1a) A1 (AO 1.1) B1 (AO 2.2a) [3]	One probability from B(100, 0.2) Both correct Correct conclusion, cwo	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
10			$f(x) = \frac{3}{x^4}$ $(x \geq 1)$	M1 (AO 1.1a) A1 (AO 1.1)	Attempt to differentiate F(x)	
			$E(\sqrt{X}) = \int_1^{\infty} \sqrt{x} f(x) dx$	M1FT (AO 1.1a)	Limits wrong or omitted: M1A0	
			$= \int_1^{\infty} 3x^{-3.5} dx$	A1FT (AO 1.1)	FT on their 3 and power of x	
			$= \frac{6}{5}$ or 1.2	A1 (AO 2.2a) [5]	Awrt 1.2 BC	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
11		i	$\int_0^4 \frac{1}{64}x^2(16-x^2)\mathrm{d}x =$ $\left[\frac{1}{64}\left(\frac{1}{3}.16x^3-\frac{1}{5}x^5\right)\right]_0^4 or \left[\frac{1}{12}x^3-\frac{1}{320}x^5\right]_0^4$ $\frac{32}{15}$ or 2.133…	M1 B1 A1 3	Attempt to integrate x f(x), correct limits *somewhere* Correct indefinite integral, aef Answer, exact or anything rounding to 2.13 Examiner's Comments For most this was straightforward, but some attempted to find the median. The misread $x(16-x)^2$ for $x(16-x^2)$ was seen several times.	Allow numerical, but if algebraic, powers must increase to score M1
		ii	(a)$\int_0^q \frac{1}{64}x(16-x^2)\mathrm{d}x = \frac{3}{4}$ $\frac{1}{64}\left(8q^2-\frac{1}{4}q^4\right)=\frac{3}{4}$ *or* $\frac{1}{8}q^2-\frac{1}{256}q^4=\frac{3}{4}$ $q^4 - 32q^2 + 192 = 0$	M1 A1 A1 3	Correct integral, limits 0, q Correct unsimplified equation Correctly obtain given equation, www Examiner's Comments Most correctly obtained the given equation; it was necessary to show the limits of 0 and q (or q and 4 if equated to ¼).	Allow limits q, 4 if equated to ¼, not otherwise Withhold if "simplified", e.g. to $-32q^6 + 192 = 0$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	(b) $q^2 = 8$ or 24 $q = \pm\sqrt{8}$ or $\pm\sqrt{24}$ $q = \sqrt{8} = 2\sqrt{2}$	M1 A1 A1 3	Solve for q^2 Acknowledge other answers (or explicitly reject) $q = \sqrt{8}$ or $2\sqrt{2}$ as sole answer, allow 2.83 as sole final answer *only if* exact also seen somewhere Examiner's Comments Almost everyone easily recognised a quadratic in q^2, but candidates at this level were expected to acknowledge four solutions (two of them negative) and then to finish with an exact solution of $2\sqrt{2}$ or $\sqrt{8}$ only. Those who went straight from the equation to 2.828 scored only 1 mark out of 3.	Numerical only: give M1 for 2.83 (2.828...) seen $\sqrt{8}$, or not $\pm$, and no other comment: M1A0A1
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
12		i	Articles must be received independently of one another and at constant average rate	B1 B1 2	At least one must be contextualised for *any* marks Independent stated, allow "probability independent" Allow "uniform" rate but not "constant" rate *Not* "probability is constant" Examiner's Comments There were many poor answers in stating the modelling assumptions for a Poisson distribution. "Randomly" was stated in the question and is therefore not an extra assumption; "singly" is as usual irrelevant. Many answers used the standard words but made no sense, suggesting that their users had no idea of what they meant — for example, "an article must occur at constant average rate". Again as usual, there were many who thought that "the probability of any article being received must be constant", regardless of the fact that this is (a) a binomial condition rather than a Poisson one, and (b) meaningless. Likewise, "the probability of an article being received is	If extras, e.g. "singly" or "randomly", then max 1 Allow "receipt of one doesn't affect receipt of another" Any implication of regularity: *can't* get second B mark

Exam Papers Practice

Question			Answer/Indicative content	Marks	Part marks and guidance	
					independent" is strictly incorrect; independence refers to events and not to probabilities. A simple correct answer was that articles should be received at a constant average rate throughout the week and independently of one another.	
		ii	Po(4.8), left-hand tail P(< 5) = 0.4763	M1 A1 2	Po(2 × 2.4), stated or implied, e.g. by 0.6510 | 0.4946 or 0.4582: M1A0 Examiner's Comments Generally very well done.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		iii	$e^{-2.4}\frac{2.4^r}{r!} = 2.5 \times e^{-2.4}\frac{2.4^{r+1}}{(r+1)!}$	M1*	One correct probability from formula	SC: T&I, or tables or calculator: 0	
				A1	Correct equation, allow from λ = 4.8 [= MR]	r + 1!: allow if used as if it were (r + 1)!	
			r + 1 = 2.5 2.4	*depM1	Simplify exp and ! correctly to linear equation in r	Wrong use of logs: M0A0	
			r = 5	A1 4	r = 5 only, www		
					Examiner's Comments A pleasing number of candidates worked this correctly. However, some were unable to deal with the algebraic manipulation, whilst others ignored the requirement to produce an algebraic solution by omitting working and using tables or a calculator for part or all of their solution.		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	Po(120) ≈ N(120, 120)	M1	Normal stated or implied, mean 50 × 2.4	
				A1	Both parameters correct, allow √ errors	
			$1-\Phi\left(\frac{139.5-120}{\sqrt{120}}\right)$	M1	Standardise, allow no / wrong cc and / or √ errors, can be implied by correct answer	If answer wrong, do not give M1 unless correct *notation* used (i.e., calculator notation such as "cdfnorm" does not qualify for M1 even if answer is recognisable)
				A1	Both cc and correct √	
			= 1 – Φ(1.78) = 0.0375	A1 5	Answer, anything rounding to 0.0375	
					Examiner's Comments Generally very well done. A large proportion of candidates got the continuity correction right.	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
13		i	B(74, 1/37) ≈ Po(2) P(≥ 4) = 1 – P(≤ 3) = 1 – 0.8571 = 0.1429 n > 50, (np =) 2 < 5	M1 M1 A1 B1 4	Po(2) stated or implied RH tail of their Poisson Answer, a.r.t. 0.143 Both conditions, dep on Poisson Allow "n large, p small" or "n large, np < 5". FT on their 74 × (1/37) if less than 5 Examiner's Comments Generally well done, although the conditions for the Poisson approximation to the binomial were often poorly stated; "np = 2 < 5" was often seen, but "n > 50", or "n is large" was also needed.	If formula used, must be correct formula, and > 1 term 0.14037 (from exact binomial): 0 np condition needs their 2 seen *somewhere* If numbers used, must be compared with 50 and 5 Extra or wrong conditions (e.g. nq): B0 *If MR leading to np > 5, please consult TL*
		ii	B(148, 18/37) ≈ N(72, 36.973)	M1 A1	N(np, ...) attempted Both parameters correct, allow	• Must see evidenc e of ap propriat e appro

Question			Answer/Indicative content	Marks	Part marks and guidance		
			$72 + 0.5 + 2.807 \times \sqrt{36.973}$ $= 89.57$ Hence $N_{min} = 90$	M1 B1 A1 A1 6	√ errors $[= \frac{1368}{37}]$ $72 + z\sigma$, allow σ^2 and / or no cc, do not allow $\sqrt{n}$ divisor $z = 2.807$ or 2.808 or 2.809 or a.r.t. 2.81, allow –ve [can be implied by 89.6 or 89.1 or 88.9] 89.6 or 89.1 seen or implied 90 only, www (although cc can be omitted if rounded up)	ximation • No working: 0 (approximation required) • 0.00196 implies exact binomial: 0 • BUT $\sqrt{37}$ may be $\sqrt{36.97}$, not $\sqrt{n}$ Ignore inequalities until final mark Allow from 89.07 and checked using exact binomial	
					Examiner's Comments Most candidates correctly used N(72, 36.97) but many then chose the wrong tail or the wrong z-value or both. A large number did not deal correctly with the continuity correction and / or the need to round up 89.56 (or 89.06) to the next whole number, 90; answers of 89.6 or 89 were common, and there was much inaccurate numerical work.		
			Total	10			

Question			Answer/Indicative content	Marks	Part marks and guidance	
14			$\frac{70-\mu}{\sigma}=0.842$; $\frac{81-\mu}{\sigma}=1.282$	M1*	Stand'ise once & equate to Φ^{-1}, allow sign/√/cc errors	"P(> 81) = P(≥ 80)" etc or "1 − 1.282" is M1A0
				A1	LHS both correct, signs consistent on both sides	Can get M1A1 even if z wrong provided they are z
				B1	Both z values, ∈ [0.841, 0.842] and [1.281, 1.282]	Can award B1 even if signs are wrong
			$11 = 0.44\,\sigma$			
			$[\sigma = 25]$ $\mu = 48.93$	dep*M1	Solve to get μ or σ	e.g. $\mu - a = 1.282\sigma$
				A1	Either, μ ∈ [48.9, 49(.0)], σ ∈ [25, 25.1], www	
			$81 - \mu = \mu - a$	M1	Equation for a, correct signs [may involve 1.282]	[a needs to be less than their μ]
			$a = 16.9$	A1 7	a in range [16.8, 17], www	
					Examiner's Comments This question on finding the mean and variance of a normal distribution began easily, as the probabilities given involved no difficult choice of signs. Most candidates got as far as finding the mean (it is not actually necessary to find the variance) but there were frequent sign mistakes in finding the value of a. Many	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					candidates were inaccurate in their working; premature rounding often lost marks by leading to final answers outside the allowed tolerances.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
15			Pdf of U=$\frac{1}{f}$	B1	$\int_{3f}^{4f}\frac{1}{f}du=1$ and attempt at int. by subn. M1 $\frac{du}{dv}=\frac{-f^2}{(v-f)^2}$ B1 $\int_{\frac{4f}{3}}^{\frac{3f}{2}}\frac{f}{(v-f)^2}dv=1$ = 1 so pdf of V is $\frac{f}{(v-f)^2}$ A1 range B1	
			$V=\frac{Uf}{U-f}$ or $U=\frac{Vf}{V-f}$	B1		
			$\int_{3f}^{u}\frac{1}{f}du=\frac{u}{f}-3$			
			cdf of $V=\mathrm{P}(\frac{fU}{U-f}\le v)$			
			1-$(\frac{vf}{v-f}-3)$ oe	B1		
			differentiate	M1		
			$\frac{f}{(v-f)^2}$	A1		
			$\frac{4f}{3}\le v\le\frac{3f}{2}$	B1		
				[6]	Examiner's Comments This was a very difficult question. Many did not realise that the pdf of U was $\frac{1}{f}$. Some that did, integrated wrt to f rather than U. Most scored the second mark. Very few scored the third mark. Most scored the fourth mark for attempting differentiation, sometimes of very strange functions. Many gave the final range the wrong way round.	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i	$\int_0^2 kx^2 dx = 0.5$ oe $= \frac{3}{16}$ AG	M1 A1 [2]	Distribution is symmetrical. Examiner's Comments Almost all the candidates answered correctly.	
		ii	$\int_{1.5}^{2} \frac{3x^2}{16} dx + \int_2^3 (\frac{3x^2}{16} - \frac{3x}{2} + 3) dx$ $\left[\frac{x^3}{16}\right]_{1.5}^{2} + \left[\frac{x^3}{16} - \frac{3x^2}{4} + 3x\right]_2^3$ $= 0.727 = \frac{93}{128}$	M1 B1 A1 [3]	ignore limits for this mark Examiner's Comments Most gained full marks. Some used a lower limit of 0 in the first integral.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\int_2^U \left(\frac{3x^2}{16} - \frac{3x}{2} + 3\right) dx = 0.25$ oe	M1		
			$U^3 - 12U^2 + 48U - 60 = 0$ oe	A1		
			$2.405 \to -0.0577$	M1	if GC need answer to 3 or more dp. 2.412598948	or Newton-Raphson or iteration $\int_{2.41}^{4} f(x)\,dx$ = 0.2512 SCB2
			$2.415 \to 0.018$			
			change of sign, $U = 2.41$ AG	A1		
				[4]	Examiner's Comments Almost all the candidates gained the first mark, and most of these went on to score the second. Many simply substituted 2.41 at this point and gained no more marks. However, there were some very good solutions particularly from those who realised that f(x) in the correct range could be written as $\frac{3}{16}(x-4)^2$. This led to the correct exact solution $4+\sqrt[3]{-4}$. Various iterations and Newton-Raphson produced correct solutions. Those who used a GC and gave the answer to more than three significant figures were also allowed full marks.	
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance		
17		a	$P(Y \le y) = P\left(\frac{1}{X^2} \le y\right)$ $= P\left(X \ge \frac{1}{\sqrt{y}}\right)$ $= 1 - F\left(\frac{1}{\sqrt{y}}\right)$ $= \begin{cases} 1 - \frac{1}{16y} & y > \frac{1}{16}, \\ 0 & \text{otherwise.} \end{cases}$	M1(AO1.1a) E1(AO2.1) M1(AO2.1) E1(AO3.1a) B1(AO1.1) [5]	Attempt to write Fy in terms of X Make X the subject 1 – F (inverse function) $1 - \frac{1}{16y}$ correct, www 0 and ranges correct (independent)	Withhold if extra range(s) given	
		b	PDF of y is $\frac{1}{16y^2}$ $\int_{\frac{1}{16}}^{\infty} \frac{y}{16y^2}\,dy$ $= \left[\frac{1}{16}\ln y\right]_{\frac{1}{16}}^{\infty}$ and ln y is undefined as $y \to \infty$	M1(AO3.1a) M1(AO1.1) A1(AO2.1) E1(AO3.2a) [4]	Differentiate CDF to find PDF of Y Multiply by y and integrate, using their limits Integration must be shown explicitly Correctly justify given statement	A0A0 For "calculator gives math error" or similar	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	OR PDF of x is $\frac{1}{8}x$ $E(Y)=E\left(\frac{1}{X^2}\right)$ $=\int_0^4 \frac{1}{x^2}\frac{1}{8}x\,dx$ $=\left[\frac{1}{8}\ln x\right]_0^4$ and ln x is undefined as $x \to 0$	M1 M1 A1 E1 [4]	Differentiate CDF to find PDF of X Integrate $\frac{1}{x^2}\times$ PDF, limits 0, 4 Integration must be shown explicitly Correctly justify given statement	A0A0 For "calculator gives math error" or similar	
			Total	9			

Question			Answer/Indicative content	Marks	Part marks and guidance		
18		a	$\mu = \int_0^\infty 0.8xe^{-0.8x}dx = 1.25$ $E(X^2) = \int_0^\infty 0.8x^2e^{-0.8x}dx\ [= 3.125]$ $Var(X) = 3.125 - 1.25^2$ $= 1.5625$	M1(AO1.1a) A1(AO1.1) M1(AO1.1) A1(AO1.1) [4]	Attempt $\int xf(x)dx$ Obtain 1.25 or exact equivalent Attempt $\int x^2f(x)dx - \mu^2$ Obtain $\frac{25}{16}$ or exact equivalent	BC or awrt 1.56 BC	
		b	$P(1 \le x < 2) = \int_1^2 0.8e^{-0.8x}dx$ = 0.247432 (6 s.f.) There are 60 specimens, so the expected frequency is 0.247432x60 =14.846	M1(AO1.1) E1(AO2.1) A1(AO1.1) E1(AO2.2a) [4]	Correct pdf Integrate between 1 and 2 Correct answer, allow 3 s.f. Multiply probability by 60 and correctly obtain given answer AG	Requires clear use of notation BC	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	H_0 : data consistent with distribution H_1 : data not consistent Combine cells to get O / E / $(O-E)^2/E$: 24 / 33.040 / 2.4734 22 / 14.846 / 3.4474 10 / 6.6707 / 1.6613 4 / 5.4431 / 0.3826 $\sum \frac{(O-E)^2}{E} = 7.965$ $\chi^2_3(0.95) = 7.815$ and $7.965 > 7.815$ $(0.95) = 7.815$ and $7.965 > 7.815$ Reject H_0. Evidence that the data is not consistent with distribution	B1(AO2.5) M1(AO1.1a) M1(AO1.1) A1(AO1.1) A1(AO3.4) B1(AO1.1) A1FT(AO2.2b) [7]	Or equivalent Combine last two cells Calculate $\frac{(O-E)^2}{E}$ for at least one cell At least two $\frac{(O-E)^2}{E}$ values correct χ^2 in range [7.96, 7.97] Comparison with 7.815 State not consistent with distribution FT on numerical errors only	B C	
			Total	15			

Question			Answer/Indicative content	Marks	Part marks and guidance		
19		a	$\sum J + \sum K \sim N(12.4, 0.0344)$ P(>12.7) =1 – 0.9471=0.0529	M1(AO1.1a) A1(AO1.1) A1(AO1.1) [3]	Consider the sum ~ N(12.4,... Standard deviation or variance correct awrt 0.053 BC	0.232 or 0.68: M1A0	
		b	$K - 0.75J \sim N(0.05, 0.003625)$ $P(>0) = \Phi(0.08305)$ = 0.7969	M1(AO1.1a) A1(AO1.1) A1(AO1.1) [3]	Or $4K - 3J \sim N(0.2, ...)$ Standard deviation or variance correct 0.0043 or 0.085: M1A0 awrt 0.797 BC		
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance	
20		i	$\int_0^1 ax^3 dx + \int_1^2 ax^2 dx = 1$	M1		
		i	$a = \frac{12}{31}$ AG	A1	Need to see $a/4$ & $7a/3$ oe	Examiner's Comments Almost all the candidates answered correctly.
		ii	$\int_0^1 ax^4 dx + \int_1^2 ax^3 dx$	M1	allow $x.ax^3$, $x.ax^2$	
		ii	$\frac{237}{155}$ or 1.53	A1		Examiner's Comments As (i).
		iii	H_0: data can be modelled by X H_1: data cannot be modelled by X oe	B1	Or f(x)	
		iii	Expected values [6, 90,] 304, 592	M1A2	M1A1 for 1 correct	eg $992 \times \frac{12}{31}\left[\frac{x^3}{3}\right]_1^{1.5}$ for M1
		iii	$TS = \frac{(8-6)^2}{6} + \ldots + \frac{(613-592)^2}{592}$	M1		
		iii	= 3.51(2)	A1		
		iii	TS < 7.815, do not reject H_0	M1	ft TS not CV. Allow 7.8	
		iii	Insufficient evidence that data cannot be modelled by X.	A1	CWO	Examiner's Comments Roughly half the candidates gave a fully correct answer. Some did not use f(x) to obtain the expected values. A common error was to use eg 281, 611 so that O-E was always 2 or −2. Some gave inadequate hypotheses, others used an incorrect critical value.
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
21		i	pdf of $R = \frac{1}{10}$ soi	B1	Method based on int. by subn. 1st B1 as main scheme. $\int_0^{10} \frac{1}{10} dr = 1$ M1* $\frac{dA}{dr} = 8\pi r$ *M1 $\int_0^{400\pi} \frac{1}{80\pi} \frac{dA}{\sqrt{\frac{A}{4\pi}}} = 1$ M1 pdf= $\frac{1}{40\sqrt{\pi a}}$ B1 attempt int to obtain cdf. M1 $\frac{\sqrt{a}}{20\sqrt{\pi}}$ A1 final B1 for range as main scheme.	NOT $8\pi r$ alone
		i	Cdf of R = $\frac{r}{10}$ allow $\frac{x}{10}$	B1		
		i	[$F_A(a)$ =] $P(A \le a) = P(4\pi R^2 \le a)$	M1		
		i	$P(R \le \sqrt{[a/4\pi]}) = F_R(\sqrt{[a/4\pi]})$	M1	allow M0M1 for eg πr^2 rearranged and substituted correctly.	
		i	$\frac{\sqrt{a}}{20\sqrt{\pi}}$ (SC B1 if obtained without M marks)	A1		
		i	$f_A(a) = \frac{1}{40\sqrt{a\pi}}$	M1A1	M1 for diffn.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$0 \le a \le 400\pi$	B1		Examiner's Comments Once again, about half the candidates gave a fully correct answer. Some candidates used an incorrect formula for the surface area of a sphere, despite this formula being in the formula booklet. Marks were also lost for using the incorrect case for R/r and A/a. Some forfeited the first two method marks and immediately substituted $\sqrt{\frac{a}{4\pi}}$ into $\frac{r}{10}$.
		ii	$\frac{\sqrt{200\pi}}{20\sqrt{\pi}}$	M1	Sub 200π in their cdf.	$4\pi r^2 = 200\pi$ $r=\sqrt{50}$ prob$=\frac{\sqrt{50}}{10}$ M1
		ii	0.707 or $\frac{1}{\sqrt{2}}$	A1		0.707 A1 Examiner's Comments Most candidates knew that they needed to substitute 200π into their CDF. Those who had the correct CDF usually obtained the correct answer.
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i	$\int_0^2 kx\,dx + \int_2^4 \frac{k}{2}(4-x)^2\,dx = 1$	M1		
		i	$2k + \frac{8k}{6}$ $(= 1)$	B1	Ignore limits for this mark.	
		i	$k = \frac{3}{10}$ AG	A1	Examiner's Comments Almost all candidates answered these parts correctly.	
		ii	$\int_0^2 kx^2\,dx + \int_2^4 \frac{kx}{2}(4-x)^2\,dx$	M1		
		ii	$\frac{1}{10}x^3, \frac{3}{20}(8x^2 - \frac{8x^3}{3} + \frac{x^4}{4})$	B1,B1	$\frac{-x(4-x)^3}{3} - \frac{(4-x)^4}{12}$ or $\frac{x^2(4-x)^2}{2} + \frac{4x^3}{3} - \frac{x^4x^4}{44}$ from int by parts. Ignore limits for these marks.	
		ii	1.8	A1	Examiner's Comments Almost all candidates answered these parts correctly.	
		iii	$\frac{3x^2}{20}$ $(0 \leq x < 2)$	B1		
		iii	"0.6"$+\frac{k}{2}\int_2^X (4-x)^2\,dx$ oe	M1	Or $-\frac{(4-x)^3}{20} + c$ and attempt to find c.	
		iii	$1 - \frac{(4-x)^3}{20}$ oe, $(2 \leq x \leq 4)$	A1	$0.05x^3 - 0.6x^2 + 2.4x - 2.2$	
		iii	0 $x < 0$, 1 $x > 4$	B1	0, 1 and all ranges. Examiner's Comments Most candidates scored full marks. The most common mistake was not to add 0.6 to $\frac{k}{2}\int_2^X (4-x)^2 dx$.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$1-\frac{(4-x)^3}{20}=0.75$	M1*		
		iv	$(4-x)^3=5$	*M1	Valid attempt to solve. Must produce soln.	$Q^3 - 12Q^2 + 48Q - 59 = 0$
		iv	2.29	A1	Examiner's Comments Candidates who used $\int(4-x)^2dx=-\frac{(4-x)^3}{3}$ did better than those who tried to solve $0.05x^3 - 0.6x^2 + 2.4x - 2.2 = 0.75$. Candidates with graphical calculators were at an advantage when using this method, but there were many correct solutions using Newton-Raphson and Iteration. Over half the candidates scored full marks.	
			Total	14		

Question			Answer/Indicative content	Marks	Part marks and guidance	
23		i	$\int_0^{\frac{1}{2}\pi} k\sin x\mathrm{d}x + \int_{\frac{1}{2}\pi}^{\pi} k(2-\frac{2x}{\pi})\mathrm{d}x = 1$	M1		
		i	$[-k\cos x]_0^{\frac{1}{2}\pi} + k\left[2x-\frac{x^2}{\pi}\right]_{\frac{1}{2}\pi}^{\pi} = 1$	B1	Both integrals correct, ignore limits.	
		i		M1	Substitute limits and attempt to simplify	
		i	$\frac{4}{4+\pi}$AG	A1	Examiner's Comments Almost always correctly answered. ∫sin x cos x was the most common error.	
		ii	$k\int_0^{\frac{1}{2}\pi} x\sin x dx + k\int_{\frac{1}{2}\pi}^{\pi} x(2-\frac{2x}{\pi})dx$	M1		
		ii	$k[-x\cos x+\sin x]_0^{\frac{1}{2}\pi} + k\left[x^2-\frac{2x^3}{3\pi}\right]_{\frac{1}{2}\pi}^{\pi}$	M1	Correct method for both integrals	Allow 1 error for M1.
		ii		A1	Both integrals correct, ignore limits.	
		ii	= 1.48 (3sf)	A1	Examiner's Comments The same error was also seen in this part. However, most candidates scored full marks.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
24		i	$A = X(10 - X)$	B1	from base x height	
		i	Use CTS	M1	or quadratic formula	Allow verification.
		i	$A = 25 - (X - 5)^2$ AG	A1	Examiner's Comments Most candidates scored full marks, but some scored only one mark for $A = 10x - x^2$. They then omitted $A = -(x^2 - 10x)$, jumping straight to the given answer, which was not acceptable.	
		ii	$f_X(x) = ½$	B1	Ignore range. Examiner's Comments Most candidates answered this part correctly.	
		iii	$F_X(x) = ½\,x$	B1	Only if (ii) correct.	
		iii	$(F_A(a) =)\ P(A \le a) = P[X(10 - X) \le a]$	M1		
		iii	$= F_X(5 - \sqrt{(25 - a)})$	A1	Fully justified.	X(or x) $\ge 5 + \sqrt{(25 - a)}$ is impossible.
		iii	$= \frac{1}{2}(5 - \sqrt{25 - a})$ AG	A1		
		iii	$0 \le A \le 16$ AG explained.	B1	eg $x = 2 \rightarrow a = 16$ Examiner's Comments The most common mark was 3 out of 5, for $F_X(x)=½x$, satisfactory explanation of the change of limits and $P(A \le a) = P(25 - (X - 5)^2 \le a)$. Most obtained $P((X - 5)^2 \ge 25 - a)$ but did not go on to say that this was equal to $P[X \ge 5+ \sqrt{(25 - a)}$ or $X \le 5 - \sqrt{(25 - a)}]$ and to explain that the first inequality was impossible, or an equivalent statement.	$F_A(16) = 1$ is not enough.

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$f_A(a) = \frac{1}{4}(25 - a)^{-\frac{1}{2}}$	M1,A1	M1 for attempt at differentiation. Examiner's Comments Almost all the candidates knew that they had to differentiate the answer given in part (iii). Some did not do this correctly, obtaining ½, −½ or −¼ instead of ¼.	
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance	
25		i	(Continuity at x = 1 gives) a = b	B1	a = b seen or implied, eg by a = 1.2, b = 1.2	
		i	$\int_0^1 ax\mathrm{d}x+\int_1^2 b(2-x)^2\mathrm{d}x=1$	M1	Even from two sep. areas eg 0.5, 0.5.	
		i	$\frac{a}{2},\left[\frac{-b}{3}(2-x)^3\right]_1^2$ (= 1) oe both seen	B1	Allow $\left[\frac{b}{3}(2-x)^3\right]_1^2$ Allow without limits $\ldots+b\left[4x-2x^2+\frac{x^3}{3}\right]_1^2$	
		i	$a/2 + b/3 = 1$	B1	cwo	
		i	Solving gives $a = b = 6/5$	A1	Examiner's Comments Most candidates scored three marks for reaching $\frac{a}{2}+\frac{b}{3}=1$. Better candidates realised that $a = b$ and went on to score full marks. Some candidates wasted time by trying to use E(X) = 1, which is wrong anyway. Almost all candidates had some idea of how to answer this question.	
		ii	$\int_0^1\frac{6}{5}\mathrm{d}x+\int_1^2\frac{6}{5}(\frac{4}{x}-4+x)\mathrm{d}x$	M1	From $\int\frac{\mathrm{f}(x)}{x}\mathrm{d}x$ ft a, b for M1A1	
		ii	= [$6x/5$ + [(6/5) (4ln$x$ – 4x + $x_2/2$)]	A1ft	Allow $a+b(4\ln 2-\frac{5}{2})$. Either their a/b or letters.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$= (24/5)\ln 2 - 9/5$ AEF or 1.53 (3SF)	A1	Examiner's Comments Most candidates who scored full marks in (i) went on to score full marks here. Many candidates with incorrect *a* and *b*, or no *a* and *b*, scored the first two marks. Some gained the first method mark and then made an error. A very small number of candidates scored zero.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
26		i	f is non-negative over [1, e]	B1		
		i	Attempt to show area, between 1 and e, = 1	M1		
		i	$\int_1^e \ln y \mathrm{d}y = y\ln y - \int \mathrm{d}y$ $= [y\ln y - y]_1^e$ $= (\mathrm{e}\ln\mathrm{e} - \mathrm{e}) - (1\ln 1 - 1) = 1$	M1	Integrate by parts. Allow 1 error.	
		i		A1	cwo Examiner's Comments Most candidates scored three marks for knowing that they needed to show $\int_1^e \ln y \, dy = 1$ and using integration by parts to do so. Most overlooked that they also needed to state that f is non-negative over [1, e]. A few candidates realised that the non-negative property was required, but made no attempt at the integral.	
		ii	F(y) = $\int_1^y \ln t \mathrm{d}t = [t\ln t - t]_1^y$	M1	Limits must be correct. If indef integral, must have '+ c' and attempt to evaluate.	
		ii	= ($y$ln$y$ – y) – (1ln1 – 1)	A1	1ln1 – 1 + c = 0 OR elne – e + c = 1	
		ii	= $y$ln$y$ – y + 1 AG, over [1, e]	A1	Needs proper justification. Examiner's Comments The candidates who gained at least three marks in (i) usually scored all three marks here. Some candidates did not score in this part, even though they knew integration by parts was required again. They either used no limits and no '+ c' or incorrect limits.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	F (2.45) = 0.745, F (2.46) = 0.754	M1	or 1 – F = 0.255, 0.246 or $y\ln y - y + 1 = 0.75$ oe	
		iii	and 0.745 < 0.75 < 0.754 and result follows	A1	Examiner's Comments Most candidates gained both marks by showing F(2.45) < 0.75 < F(2.46). Some candidates did not realise that this is all that was required. Many of these candidates scored a method mark for $Q_3 \ln Q_3 - Q_3 + 1 = 0.75$ and then tried to rearrange this. Many gave up, some realised that they needed to use a method similar to the standard solution and a very small minority produced the correct answer 2.455(11..) by using their GC, iteration or Newton-Raphson.	
		iv	$G(x) = P(X < x)$			
		iv	$= P(\ln Y < x)$			
		iv	$= P(Y < e^x)$	M1		
		iv	$= F(e^x)$	M1	Allow M0M1A1B1.	
		iv	$= (e^x \ln e^x - e^x + 1)$ $xe^x - e^x + 1$; over [0,1]	A1;B1	Examiner's Comments Most candidates gained at least three marks. Many of these lost the first method mark for eg P(lny < x) etc, but they knew that F (e^x) was needed. Some did not produce $0 \le x \le 1$.	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
27		i	$F(x) = x^{3/2}$ $(0 < x \le 1)$	B1		
		i	$G(y) = P(Y \le y)$ $= P(1/\sqrt{X} \le y)$ $= P(X \ge 1/y^2)$	M1		
		i	$= 1 - F(1/y^2)$	M1		
		i	$= 1 - 1/y^3$ for $y \le 1$	A1,B1	B1 for $y \ge 1$ seen anywhere.	
		i	$(G(y) = 0$ otherwise) $g(y) = G'(y) = 3/y^4$ $y \ge 1$, AG (= 0 otherwise)	M1A1	M1 for differentiating Allow from $-1/y^3$ Examiner's Comments Most candidates gained most of the marks in this part. The most common error was obtaining an incorrect range for y. In recent series candidates have learned to answer this type of question well, understanding the difference between Y and y.	
		ii	(α) $\int_1^{\infty} \frac{3}{y^2}\,dy = 3$	M1A1	Ft range from (i) for M1, but not $\int_0^1 \frac{3}{y^2}dy$	
		ii	(β) $\int_0^1 \frac{3}{2\sqrt{x}}\,dx = 3$	M1A1	Examiner's Comments Those who tried to evaluate $\int_0^1 \frac{3}{2\sqrt{x}}dx$ usually scored better than those using $\int y^2 g(y)\,dy$, many having the wrong limits obtained in (i).	
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
28		i	[x represents a] possible value(s) taken by X	B1	Must refer to, or imply, both x and X or “the random variable” Ignore extra unless definitely wrong Examiner's Comments Many candidates have learnt that x is a value taken by the random variable X, though some still seem to have no clear idea of what is happening and refer to “inputs into the probability function” or some such. If this question has focussed on such a widespread misunderstanding and helped to reduce it, it will have served a good purpose.	
		ii	$\int_2^\infty ax^{-3}+bx^{-4}\mathrm{d}x=\left[-\frac{a}{2x^2}-\frac{b}{3x^3}\right]_2^\infty=\frac{a}{8}+\frac{b}{24}$	B1	Correct indefinite integral [from any set of limits or none]	
		ii		M1	Integrate and substitute limits to obtain one expression	
		ii	*or* $\int_1^\infty ax^{-3}+bx^{-4}\mathrm{d}x=\left[-\frac{a}{2x^2}-\frac{b}{3x^3}\right]_1^\infty=\frac{a}{2}+\frac{b}{3}$	M1	Integrate and substitute limits to obtain a second expression The limits must be two of (1, ∞), (1, 2) or (2, ∞), allow (3, ∞) for “⩾ 2”	
		ii	*or* $\int_1^2 ax^{-3}+bx^{-4}\mathrm{d}x=\left[-\frac{a}{2x^2}-\frac{b}{3x^3}\right]_1^2=\frac{3a}{8}+\frac{7b}{24}$	M1	Equate two expressions from definite integrals to 1 or $\frac{3}{16}$ or $\frac{3}{16}$ appropriate, and attempt to solve	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\frac{a}{2}+\frac{b}{3}=1$ or $\frac{a}{8}+\frac{b}{24}=\frac{3}{16}$ or $\frac{3a}{8}+\frac{7b}{24}=\frac{13}{16}$	A1	Both equations correct, any equivalent simplified form, can be implied [“simplified” = one a term, one b term, one number term]	
		ii	Solve to get	A1	Correctly show a = 1 AG, www	
		ii	a = 1 $b=\frac{3}{2}$	B1	Correct value of b obtained from at least one correct equation SC: One equation only: M1B1 M0M0A0 A0B1, max 3/7 Two equations, assume a = 1, solve for b, checked in other equation: 7/7 Examiner's Comments Candidates made surprisingly heavy weather of this question (many scripts stretched to several additional pages), but it is simply the continuous equivalent of an often-asked S1 question. The intention was that candidates should use the given information to integrate the PDF between 2 and ∞ to obtain one equation, such as $\frac{a}{8}+\frac{b}{24}=\frac{3}{16}$, and use the result that the total probability is 1 to obtain a second equation such as $\frac{a}{2}+\frac{b}{3}=1$. However, quite a lot of candidates didn’t think of this second condition, although many used the equivalent method of finding the area between	

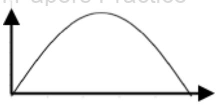
Question			Answer/Indicative content	Marks	Part marks and guidance	
					1 and 2. Problems that arose included attempts to use 0 as a limit or thinking that P(≥ 2) means P(⩾ 3). As the question said "show that *a* = 1" it was not sufficient to produce two simultaneous equations and just write down the answers, perhaps from a calculator. Enough working was needed to get to the point where the value *a* = 1 was obvious, unless the candidate found b first. Some assumed that *a* was 1 and could get 3 marks out of 7 if they then obtained the correct value of *b* from a correct equation. *b* = 11/2.	
		iii	$\int_1^\infty ax^{-2}+bx^{-3}\mathrm{d}x=\left[-\frac{a}{x}-\frac{b}{2x^2}\right]_1^\infty$	M1	Integrate $x\mathrm{f}(x)$, limits 1 and ∞ seen somewhere	
		iii	$\left\{=a+\frac{b}{2}\right\}$	B1ft	Correct indefinite integral, their *b*, can be implied by correct answer *Expect to see* $\int_1^\infty x^{-2}+\frac{3}{2}x^{-3}\mathrm{d}x=\left[-\frac{1}{x}-\frac{3}{4x^2}\right]_1^\infty$	
		iii	$=1\frac{3}{4}$	A1	Correctly obtain ¾ or a.r.t. 1.75 www, allow from calculator Examiner's Comments Many candidates obtained full marks on this question, and even if they had obtained the wrong value of *b* in part (ii) could score 2/3. There were pleasingly few mistakes with the algebra. 1.75.	
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance	
29		i	$\int_{-3}^{3} \frac{3}{2a^3} x^2 dx = \left[\frac{x^3}{2a^3}\right]_{-3}^{3} = \frac{27}{a^3}$	M1dep*	Integrate, attempt at correct seen limits *somewhere*	Allow e.g. "< 3" = "≤ −4"
		i		B1	Correct indefinite integral, can be implied by, e.g. $27/a^3$	Allow also for a^3 on top
		i	=0.125	*M1	Equate, with limits, to 0.125 and solve	
		i	so a = 6	A1	Solve to get a = 6 exactly Examiner's Comments Most knew what to do here but there were frequent algebraic errors. Very common was a tendency for the a^3 factor to migrate from the bottom of the fraction to the top. Weaker candidates omitted the limits altogether.	Allow 6.00 but no other decimals. *Not* ±6
		ii	$\mu = 0$	B1	Stated somewhere or calculated, any a	If $\mu = 0$ not mentioned anywhere, or "$-\mu$"
		ii	$\int_{-a}^{a} kx^4 dx = \left[k\frac{x^5}{5}\right]_{-a}^{a} = \frac{3a^2}{5}$	M1dep*	Attempt to integrate $x^2f(x)$, limits $\pm a$	stated [instead of "$-\mu^2$"], B0 but can get
		ii		B1	Or exact equivalent, can be implied	remaining 4/5
		ii		*M1	Equate to 1.35 and solve	Don't need explicit $-\mu^2$ here
		ii	1.35 so a = 1.5	A1	a = 1.5 ± 0.005, allow ±1.5, ignore "must be positive"	NB: a = 3 is *not* MR but can get B1 for $\mu = 0$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii			Examiner's Comments This question had a number of poor answers. A number of candidates used the same limits (3 and −3) that they had just used in part (i). Many candidates forgot all about the mean, or failed to square it. As has been observed before, the attempt to use a single massive formula for the variance, including the square of the integral for the mean, is beyond most candidates and is not recommended. Those who attempted to find the mean by integration often got it wrong; they should have been aware that $\mu = 0$ by symmetry, even if they were not confident enough to use 0 straight away. Various algebraic errors, such as the migration of a^3 mentioned in part (i), also hindered candidates.	

Exam Papers Practice

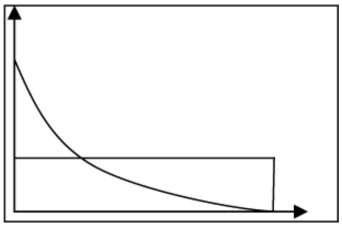
Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	x is a value [values] that X takes	B1	Ignore irrelevancies or extra wrong, unless contradictory Examiner's Comments This question confirmed the suspicion that only a small proportion of candidates have a clear idea as to what a probability density function is. The simple, correct answer, x is a value taken by X, was less often seen than answers to questions that have been asked in the past, such as 'values at the extremities are more likely than those at the centre'. A statement such as 'X is a continuous random variable' seems to cause problems; candidates seem to think that X is something that does or does not happen, dependent upon the value of x. It might be helpful to focus on the idea that X is the output of a process – when the variable X is observed, what comes out is a number, one particular value of which is denoted by x.	*Not* answers just about the *function*
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
30		i	$\int_0^1 \frac{\pi}{2}\sin(\pi x)\,dx = \left[-\frac{1}{2}\cos(\pi x)\right]_0^1 = \frac{1}{2} - \left(-\frac{1}{2}\right) = 1$	M1	Attempt to integrate f(x), limits (0, 1) somewhere, evidence e.g. "from calculator"	
		i		B1	Correctly integrate sin(πx) to −½cos(πx)	
		i		A1	Fully correct, need to see −½cos(πx) and final 1, no wrong working seen	
		i	and function non-negative for all x in range	B1	Non-negative asserted explicitly, allow positive or equivalent. Not just graph drawn. *(Most will not get this mark!)* Examiner's Comments A large number of candidates could integrate sin(πx) correctly, even if they found that some subsequent adjustment of the negative sign or the factor of π was necessary, and most could demonstrate that the area under the curve was 1. Almost all, however, omitted the other condition for a function to be a PDF: it must be non-negative for all values of x.	
		ii		M1	Correct shape, through 0, allow below axis outside range. Allow partial curve if clearly part of sine curve.	
		ii		A1	Fully correct including no extension beyond [0, 1]. Don't worry about grads at ends. Ignore labelling of axes	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$E(X) = ½$	B1	½ or 0.5, needs to be simplified, no working needed, *no* ft Examiner's Comments Most knew the correct shape of the graph, though some drew either not enough of a sine wave or too much. Some candidates tried to use integration in order to find $E(X)$, some realising later that they could and should simply have written the answer down. A common mistake was to give the y-coordinate of the maximum, $\pi/2$, instead of the x-coordinate, ½.	
		iii	$\int_q^1 \frac{1}{2}\pi\sin(\pi x)\mathrm{d}x = 0.75$; $\{\left[-\frac{1}{2}\cos(\pi x)\right]_q^1 = 0.75\}$	M1	Equate integral to correct probability, correct limits *somewhere* allow complementary probability (= 0.25) only if limits $(0, q)$	
		iii	$\cos(\pi q) = 0.5$	A1	$\cos(\pi q) = 0.5$ or exact equivalent	
		iii	Solve to get $q = ⅓$	A1	$q = ⅓$ or a.r.t. 0.333. SR: Numerical (no working needed): 0.333 B3, 0.33 B2 Examiner's Comments The common mistake here was to equate the integral between 0 and q to 0.75, rather than the integral between q and 1. Otherwise many correct answers were seen.	
		iv	$\int_0^1 \frac{\pi}{2}x^2\sin(\pi x)\,\mathrm{d}x - \left(\frac{1}{2}\right)^2$	M1	Integral part correct, allow limits omitted, ignore dx	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv		A1ft	Subtract their $[E(X)]^2$, allow μ in form of integral, correct limits needed, not just "μ^2" {note for scoris zoning — (ii) needs to be visible here} Examiner's Comments Most candidates scored the first mark; some lost the second one by omitting either the limits of the integral, or the value of $[E(X)]^2$.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	Values of x in range close to $\mathrm{E}(X)$ are more likely than those further away	B1	Need to see "values of x" or equivalent, and probably not "occur" *Not* "the probability of x is greater when x is close to $\mathrm{E}(X)$" etc. *Not* "PDF greater …" Examiner's Comments As usual this type of question revealed severe misunderstandings as to the nature of a PDF. Many candidates clearly think that X and x represent completely different things, a misunderstanding that can perhaps be summed up by a typical (wrong) expansion of the mis-statement in the question: "Whether or not X happens depends on how close to ½ the value of x is". Examiners were looking for evidence that candidates understood that X takes numerical values, and/or that x merely represents a possible value that X can take. Centres might find it advisable to focus on the two facts that (i) X stands for a number, not an event; and (ii) x represents merely a number that X could be.	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
31		i		M1	Upwards parabola, not below x–axis	*[scales / annotations not needed]*
		i		A1	Correct place, not extending beyond limits, ignore pointed at a	Touching axes (not asymptotic)
		i		B1	Horizontal straight line, not beyond limits, y–intercept below curve (unless curve makes this meaningless) Examiner's Comments As usual, there were many who wrongly drew the graphs beyond the limits of [0, a]. Many could not put the parabola in the right place.	Don't need vertical lines i.e., 3/3 only if wholly right
		ii	$\int_0^a \frac{3}{a^3}x(x-a)^2\,dx$	M1	Attempt this integral, correct limits seen somewhere	
		ii	$=\int_0^a \frac{3}{a^3}(x^3-2ax^2+a^2x)\,dx$	M1	Method for $\int x\mathrm{f}(x)$, e.g. multiply out or parts, independent of first M1	Multiplication: needs 3 terms
		ii		A1	Correct form for integration, e.g. multiplied out correctly, or correct first stage of parts	E.g. $\frac{3}{a^3}x\frac{(x-a)^3}{3}-\int\frac{3}{a^3}\frac{(x-a)^3}{3}\,dx$
		ii	$=\left[\frac{3}{a^3}\left(\frac{x^4}{4}-\frac{2ax^3}{3}+\frac{a^2x^2}{2}\right)\right]_0^a$	B1	Correct indefinite integral	E.g. $\frac{3}{a^3}x\frac{(x-a)^3}{3}-\frac{3}{a^3}\frac{(x-a)^4}{12}$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$= \frac{a}{4}$	A1	$\frac{a}{4}$ or exact equivalent (e.g. $0.25a$) only Examiner's Comments It was surprising to find so many S2 candidates who could not multiply out $(x - a)^2$. Commonly seen were $x^2 - 2a + a^2$ and $x^2 - 2ax + a$. Most at least knew that to integrate $x(x - a)^2$ they had to multiply out (or use parts). There were other frequent algebraic errors, too.	Limits not seen anywhere: can get M0M1A0B1A0
		iii iii	S is concentrated more towards 0 Therefore T has bigger variance	M1 A1	Reason that shows understanding of PDF Correct conclusion Examiner's Comments This question brought out all too clearly how few candidates understand pdfs. A correct answer is that T is equally likely to take any value in the range, whereas S is more likely to take values towards 0, so T has the greater variance. The most common wrong answer was that "T is constant so has small variance" (few committed themselves to saying "zero variance"). There is a recurring misconception here. The *input* values (x) for the pdf represent the possible values taken by the random variable; the *output* values (y, or f(x)) indicate the likelihood that the random variable takes	*Not*, e.g., "T is constant"

Question			Answer/Indicative content	Marks	Part marks and guidance	
					those values. The default reaction of the lower scoring candidates is that the output values are the possible values taken by the random variable. Some make explicit their misconception that x represents some sort of input parameter, which apparently determines whether or not S or T "occurs" – as if S or T were "events" and not variables representing random numbers. Alternatively, some candidates probably think that "variance" measures variability in probability and not in the values of the random variable.	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
32			H_0: μ= 38.4 [Allow E(X) both times] H_1: μ ≠38.4	B2	Both correct: B2. One error e.g. no or different symbols, one-tail etc, B1	But $\bar{x}$, x, t etc B0. E.g. H_0: μ_0= 38.4, H_1: μ_1 ≠ 38.4: B1
			$\hat{\mu} = \bar{x} = 36.68$	B1	36.68 seen anywhere	H_0: μ= 36.68, H_1: μ ≠ 36.68: B0B0B1 *See below and exemplars*
			$\hat{\sigma}^2 = \frac{50}{49}\left(\frac{70027.37}{50} - 36.68^2\right) = 56.25$	M1	Use biased variance formula [55.125]	Single formula: M2 or M0. If M0, a
				M1	Multiply by 50/49	divisor of 49 seen anywhere gets M1
				A1	56.25	Allow rounded if clearly correct
			α: $z = \frac{36.68 - 38.4}{\sqrt{56.25/50}} = -1.62$	M1	Standardise using √50 or 50	If 50 missing, no more marks
				A1	z, a.r.t. – 1.62 or p = 0.0525	p in range [0.052, 0.053]
			> – 2.576 [*or* 0.0525 > .005]	A1ft	Compare – z with – 2.576 or + z with 2.576	Ft on z. Or p explicitly with 0.005
			β: CV is $38.4 - 2.576\sqrt{\frac{56.25}{50}} = 35.6677$	M1	CV 38.4 – $z\sigma/\sqrt{50}$, ignore 38.4 + anything	36.68 + $z\sigma/\sqrt{}$ 50: M1A0A0, M0A0
				A1	A.r.t 35.7	
			36.68 > 35.6677	A1ft	CV ft and correct comparison	Ft on wrong z or on √ only
			Do not reject H_0.	M1	Correct first conclusion, needs correct method & comparison if seen	Like-with-like, needs μ and $\bar{x}$ right way round, needs 50
			Insufficient evidence of a change in crop yield	A1ft	Contextualised, "evidence" somewhere *Not* "evidence of no change"	Ft on wrong TS and / or CV

Question			Answer/Indicative content	Marks	Part marks and guidance	
					Biased variance [55.125; – 1.638 or 0.0508] can get B2B1 M1M0A0 M1A0A1M1A1 (max 8) σ^2 used [– 1.529 or 0.0632, or – 0.12162 or 0.4144]: B2B1 M1M1A1 M1A0A1M1A1 (max 10) No $\sqrt{50}$ [– 0.2293 or 0.4092]: B2B1 M1M1A1M0 (max 6) H_0 / H_1 in terms of 36.68: can get last 4 marks *only* if (36.68 – 38.4) seen, and not (38.4 – 36.68)	Examiner's Comments This question requiring a hypothesis test for the mean μ of a normal distribution, based on an unbiased estimate of the population variance, was probably the best done on the entire paper – and of course it was the most similar to questions that have been asked before. Many scored most or all of the marks. However, one error that led to substantial loss of marks was omission of the $\sqrt{50}$ in standardising. Previous reports have drawn attention to the serious misunderstanding represented by statements such as $\bar{X} \sim N(36.68, 56.25 / 50)$. 36.68 is the sample mean and not the population mean; the underlying basis of the test is the assumption that μ is the hypothesised value (38.4) and not the sample mean. Likewise it is important to get the standardisation the right way round: $\frac{36.68-38.4}{\sqrt{56.25/50}}$ and not $\frac{38.4-36.68}{\sqrt{56.25/50}}$.
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance	
33		i	Values taken by X	B1	This answer only Examiner's Comments This simple request revealed many misunderstandings. It is clear that the letter x had no meaning in this context for many candidates. The correct answer is that x represents a value, or values, taken by the random variable X.	Not "values taken by f"
		ii	$\int_0^a kx dx = 1 \Rightarrow k = \frac{2}{a^2}$	M1	Use definite integral and equate to 1,	*Or* clear argument from triangle area
		ii		A1	Correctly obtain $2 / a^2$ Examiner's Comments Almost always correct.	
		iii	$\int_0^a kx^2 dx = \left[k\frac{x^3}{3}\right]_0^a = \frac{2}{3}a$	M1	Attempt to integrate xf(x), limits 0 and a	
		iii		B1	Correct indefinite integral seen	either here or for x^2f(x)
		iii		A1 √	Correct mean *or* correct E(X^2) [= a^2 / 2], √ on k	Can be in terms of k
		iii	$\int_0^a kx^3 dx = \left[k\frac{x^4}{4}\right]_0^a = \frac{a^2}{2}$	M1*	Attempt to integrate x^2f(x), limits 0, a	
		iii	$\frac{a^2}{2} - \left(\frac{2}{3}a\right)^2 = \frac{1}{18}a^2$	depM1	Subtract their μ^2	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		A1	Correct final answer, ae exact f, no k now Examiner's Comments Generally done very well, but some failed to square the mean (even when they had written down a formula involving μ^2), and some omitted it altogether. Examiners find that candidates who try to use a single formula, involving the subtraction of the square of an integral, are more likely to get the wrong answer; this formula seems to be too complicated for the majority of candidates. A number of candidates failed to realise that the final answer had to be given in terms of a only, and not k.	Or decimal, $0.056a^2$ or better
			Total	9		